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"The purpose of computing is insight, not numbers" – Richard Hamming

1 Field of Expertise

My field of expertise lies at the intersection of three principal fields of scientific computing -

- High Performance Computing (HPC)
- Data Visualization
- Data Analysis and Machine Learning

A brief definition of each of these terms is provided below:

High Performance Computing (HPC): High Performance Computing refers to the practice of aggregating computing power in a way that delivers much higher performance than one could get out of a typical desktop computer or workstation in order to solve large problems in science, engineering, or business. High Performance Computing is usually associated with a High Performance Computer Hardware called a Supercomputer. A Supercomputer is an exotic machine built with special technology, and at a large scale (10,000 processors) to solve some of mankind's biggest problems in science and engineering (eg. Predicting cyclones, building efficient combustion engines etc). Supercomputers are usually quite expensive, costing upwards of about $20M each.

Data Analysis and Machine Learning: Data analysis, especially in the context of science and engineering, is a process of cleaning, transforming, and summarizing scientific data with the goal of making scientific discoveries, informing scientific conclusions, and supporting decision-making. Machine Learning is a collection of techniques where computer programs automatically learn to identify patterns and structure in new data using observations, rewards, or labeled examples.

Data Visualization: Data Visualization is the representation and presentation of data, in a visual format, to facilitate understanding. Specifically, the purpose of visualization is to get insight, by means of interactive graphics, into various aspects related to some process we are interested in, such as a scientific simulation or a real-world process. Visualization allows us to

- Immediately comprehend huge amounts of data
- Facilitiate understating of both large scale and small scale features of the data
- Facilitate scientific hypothesis formation

Gaining expertise in the above fields requires the following skills:

1. **Working with Supercomputers:** Creating analysis and visualization workflows on Shaheen-II, the world's 32nd fastest supercomputer at KAUST.
2. **Building and working with large-scale, high-performance visualization facilities:** I am a founding member of the Visualization Core Lab at KAUST. Such facilities are unique, and few in the world [DeFanti et al., 2011].
3. **Other skill areas:**
    - Scientific Visualization
    - Machine Learning, Math, and Statistics
    - 3D Computer Graphics and Animation



Figure 1: Facilities at my lab - The Visualization Core Lab. Left: Tiled-Display Wall, Middle: Fully-Immersive Virtual Reality CAVE, Right: Semi-Immersive Virtual Reality Cubes.

- Software Engineering
- Domain Knowledge in scientific areas such as Climate Science, Computational Fluid Dynamics, Marine Ecology etc.

The above areas of expertise and skills make me a unique professional that supports and enables scientific discoveries, while working with domain scientists.

2 Discoveries and Contributions

My lab's primary role is to support domain scientists with our expertise in enabling discoveries in their specific research areas. My lab and I have contributed to the following scientific fields:

- Climate Science and Oceanography
- Combustion Physics
- Marine Ecology
- Material Science
- Computational Fluid Dynamics
- Neuroscience
- Applied Statistics

I supervise a staff of 6 scientists, and manage high-end visualization facilities in the lab (Figure 1). My contributions have been both direct and indirect. In each of the direct contributions (Sections 2.1 to 2.7), I have collaborated with a domain expert directly to enable discoveries in their research. My indirect contributions (Section 2.8) are through the impact created by my staff and the use of large-scale visualization facilities in my lab. The following sections outline both.

2.1 Using Data Science to Reveal Patterns in Marine Animal movements

Research Context

Marine ecologists have been maintaining a growing database of tracked animal movements over large areas and long periods of time (eg. www.movebank.org). Specifically, distinguished KAUST marine researcher Dr. Carlos Duarte and his international collaborators have meticulously tracked movements of 321 individual Northern Elephant Seals, consisting of over 1.2 million GPS points. These animals are known to perform coordinated voyages, and communicated over 5000 Kms. There are several challenges in understanding large and complex behaviors of such animals:

- User-friendly querying and visualization of these GPS tracks.

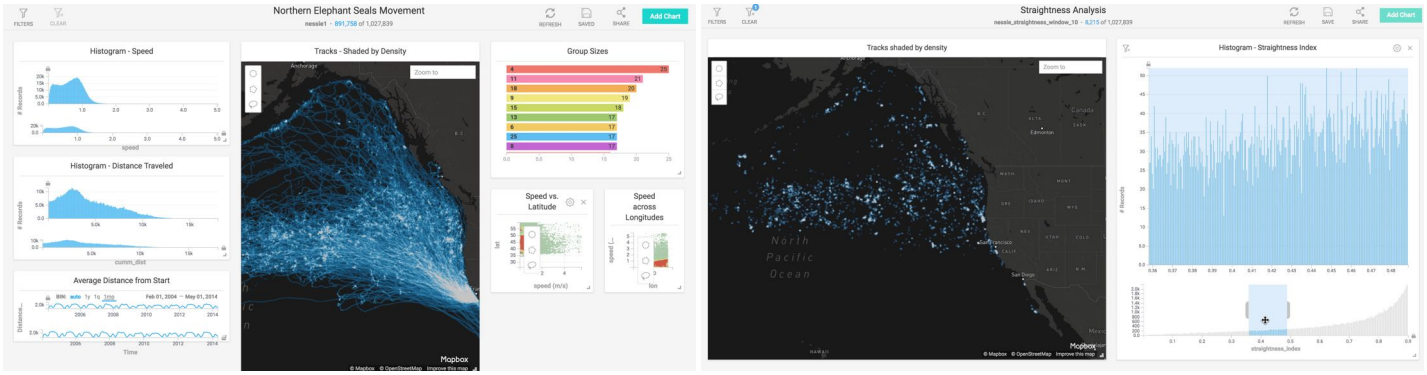


Figure 2: Left: Interactive Visualization and Analysis of Elephant Seal Tracks. Right: Discovery of foraging regions for the seals, in the Pacific Ocean.

- Identify emergent patterns from coordinated movements
- Identify interactions with climate and environment

Contributions and Impact

I collaborated with Dr. Duarte and Dr. Dan Costa from University of California, Santa Cruz, to:

- develop a data science algorithm to discover additional information from raw GPS tracks of 321 Northern Elephant Seals.
- develop an integrated visualization and data science tool to visualize and interactively analyze the nature of these Elephant Seal Movements (see Figure 2). The technique uses specialized hardware (GPUs) to perform these analyses quickly.
  [Click here for video]
- to help discover (through my tool) foraging regions in the Pacific Ocean, for these elephant seals. This confirms a previously published hypothesis in the field.
  [Click here for video]

Publications

I am a co-author in the following publication:

- Sonification of Animal Tracks as an Alternative Representation of Multi-Dimensional Data: A Northern Elephant Seal Example [Duarte et al., 2018]

Articles

- Translating elephant seal data into a symphony provides surprising insights
- Synchronized swimming for seal migrations

Videos

- Discovering the elephant seals foraging behavior
- Visualization and Analysis Tool for the Elephant Seal Tracks

2.2 Visualization and Analysis of Large Scale Combustion Simulations

Research Context

Direct Numeric Simulations of Premixed Turbulent Combustion of Methane and Air models the combustion dynamics of large combustion devices such as gas turbines etc. The numerical simulation performed by KAUST researchers [Attili et al., 2016] on the KAUST

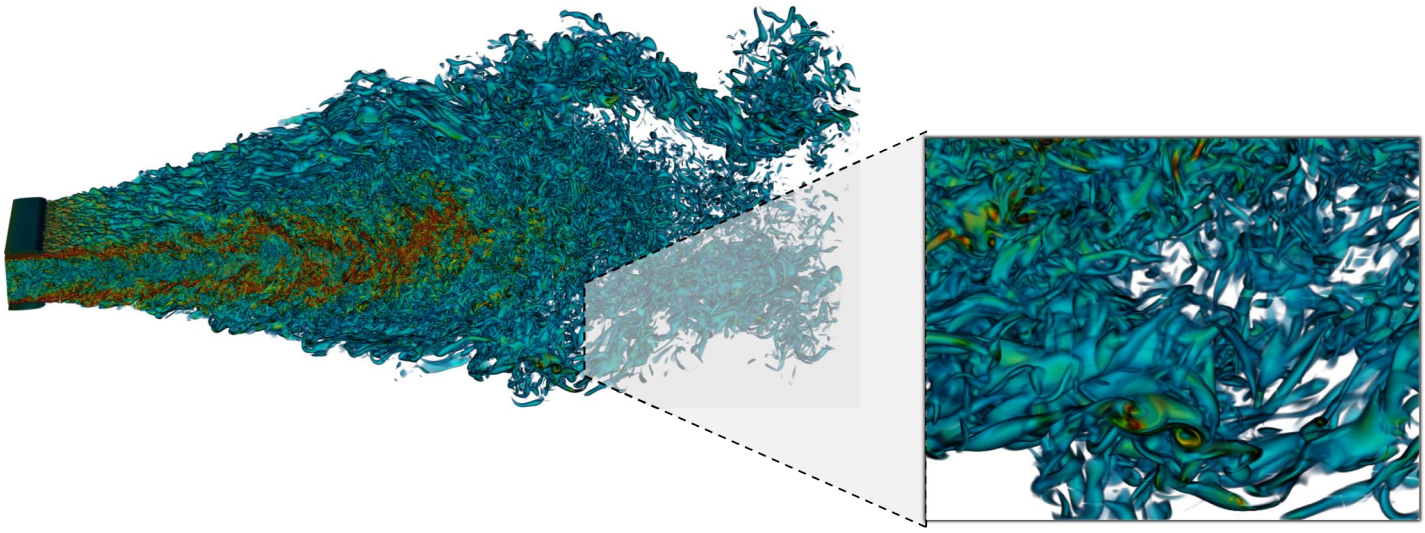


Figure 3: Visualizing turbulence structures in a 22-billion-grid point combustion simulation [Attili et al., 2016]

Supercomputing resources calculates data for 16 different variables, over a 22-billion grid points, generating about 4.3 terabytes of data per time-step (and there are over 1000 time steps per simulation). This was the largest ever DNS simulation performed (in 2017).

These simulations feature turbulences (also called vorticities) at multiple scales (see Figure 3). Turbulence affects the quality of combustion, and therefore the efficiency of these large combustion devices. The challenge therefore is to understand the physics of these turbulence structures, and to perform an interactive, qualitative and quantitative exploration of a 4.3 terabyte dataset. This is a non-trivial task, as the data has to be distributed across a supercomputer, and the results streamed to the researcher's workstation interactively.

Contributions and Impact

With over 4 Terabytes of data per timestep, and hundreds of timesteps computed, we worked with the domain scientists to create a meaningful visualization workflow that scales with such large data sizes. Our domain scientists used several hundred cores of the Shaheen II supercomputer, a high-throughput 40-Gigabit network link between the Visualization and Supercomputing Core Labs, and a high-performance 16-node GPU cluster to produce a parallel visualization technique that can also be interactive, for large datasets. In summary, my contributions were as follows:

- Created an efficient workflow for Interactive analysis and visualization of the largest ever combustion simulation, consisting of 22-billion data points, on the KAUST Supercomputer (Shaheen II).
  [Click here for video]
- Created a novel visualization streaming workflow for on-demand exploration for combustion datasets from a supercomputer, as outlined in the paper - Query-driven parallel exploration of large datasets [Atanasov et al., 2012].
- Supervised two graduate students to implement two innovative approaches to visualize combustion datasets [Binyahib, 2013] [Nagoor, 2014] [Nagoor et al., 2015].

Publications

My contributions were used in the following publication by the domain scientist:

- Large scale Direct Numerical Simulation of premixed turbulent jet flames at high Reynolds number [Attili et al., 2016].

I am a co-author in the following publications:

- Query-driven parallel exploration of large datasets [Atanasov et al., 2012].

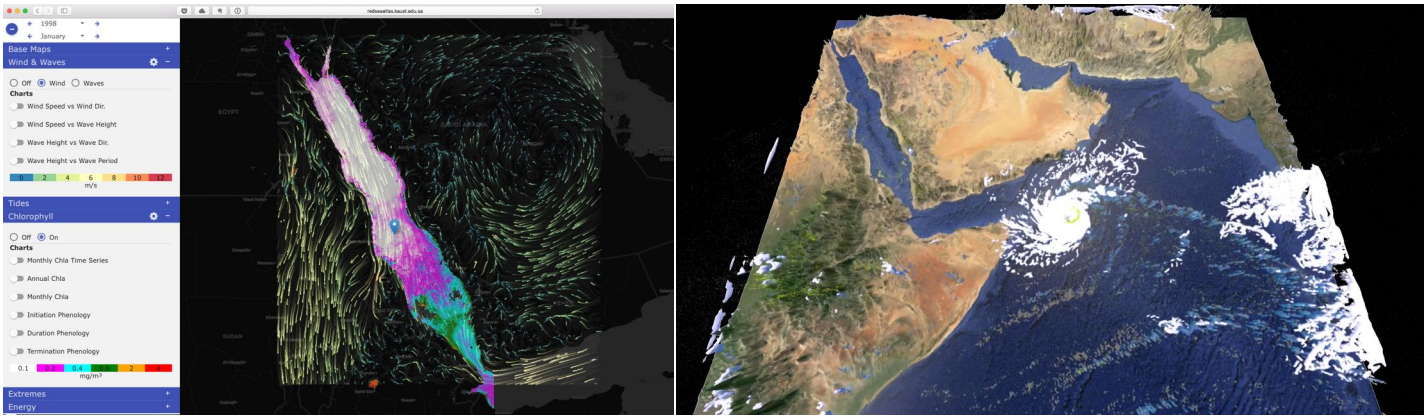


Figure 4: Left: The *RedSeaAtlas* visual-analytics tool. Right: Visualization of extreme weather events (cyclones)

- PathlinesExplorer - Image-based Exploration of Large-Scale Pathline Fields [Nagoor et al., 2015].

I supervised the following work done by M.S. students working in this research area:

- Image-Based Exploration Of Large-Scale Pathline Fields [Nagoor, 2014]
- Image-Based Exploration Of Iso-Surfaces For Large Multi-Variable Datasets Using Parameter Space [Binyahib, 2013]

Articles

- Visualizing the largest combustion simulation ever performed.

Videos

- Visualizing a Combustion Dataset
- Visualizing Streamlines (from Student Thesis Work)

2.3 Visualization and Analysis of Climate Datasets

Research Context

Climate scientists at KAUST working with ocean and atmospheric datasets deal with diverse data analysis tasks related to a variety of application areas such as tracking fisheries movements, coastal developments, environmental planning, optimization of renewable energy generation, navigational guidance for marine vessels, tourism planning, and policy-making. They encounter data from heterogeneous data sources either generated by numerical simulation models or observational data, and they often have to combine these disparate data for simultaneous analysis. These diverse data and task requirements pose great visualization challenges that are necessary to address, to enabled discoveries in climate research.

Contributions and Impact

My lab and I have been collaborating with our climate researcher, Dr. Ibrahim Hoteit, to design several custom and task-specific visual-analytics tools intended for solving problems in the ocean and atmospheric domain:

- Red Sea Atlas: We designed a custom visual analytics tool (called the *RedSeaAtlas*) intended for use by the ocean and atmospheric scientists working on different analysis tasks, dealing with multivariate ocean and atmospheric datasets for the region's Red Sea [Afzal et al., 2019] The tool has proven to be useful to our climate researchers for:
  - Tracking the flow of nutrients in the Red Sea. These nutrients are essential for marine life as they serve as a food source, and therefore identify areas most profitable for fishing.

- Analyzing extreme events in the Red Sea. These events are indicative of marine heat waves, and can cause coral bleaching if the heat-waves sustain for long periods.
- Understanding the spatiotemporal climate patterns related to wind energy in the region. These patterns are essential to optimize the locations for wind-based renewable energy infrastructure. These insights also help experts estimate extractable energy from wind at different places and then optimize the grid systems for energy generations.

[Click here for video]

- Visualizing Cyclone Chapala: The visualization in Figure 4 shows the eye of the cyclone "chapala" which formed south of the arabian peninsula in October 2015, causing significant damage to life and property in the peninsula. The cyclone was a category 4 storm with 155 mph winds, and was the second strongest tropical cyclone on record in the Arabian Sea. Over the Arabian Sea, such well structured and matured cyclones are unusual in post-monsoon periods.

  We worked with Climate Scientists (Dr. Ibrahim Hoteit) at KAUST to simulate, analyze and visualize the movements of cyclone Chapala on Shaheen II (KAUST Supercomputer). The simulation output results in several hundred gigabytes of data. We analyze and visualize these data using state of the art computational and visualization software for representing different characteristics of Chapala and to verify the accuracy of the simulation model [Theußl et al., 2016].

  [Click here for video]

Publications

I am a co-author in the following publications:

- Simulation and Visualization of the Cyclonic Storm Chapala over the Arabian Sea: A Case Study [Theußl et al., 2016]
- RedSeaAtlas: A Visual Analytics Tool for Spatiotemporal Multivariate Red Sea Data [Afzal et al., 2019]

Articles

- Seeing the Red Sea in New Ways

Videos

- RedSeaAtlas in action
- Visualizing Cyclone Chapala

2.4 KAUST-EPFL Alliance for Neuro-Inspired High Performance Computing

Research Context

The KAUST-EPFL Alliance for Neuro-inspired high performance computing was established in 2013 between Prof. Henry Markram's Blue Brain Lab at EPFL - École polytechnique fédérale de Lausanne, KAUST's Dean of Biological Sciences and Environment division - Dr. Pierre Magistretti, and my lab. The goal of this alliance was to develop biologically detailed 3D models of the blood supply to the brain, and study the glial cells coupling neurons and vasculature in the neuro-vascular-glial unit. These detailed models of the brain will facilitate understanding on how to perform computations with minimal energy requirements and with high reliability, on future neuro-inspired supercomputing systems. This will enable simulation, visualization and analysis to be performed at the same time on the same machine, interactively, laying the foundation for exploiting supercomputers for science and engineering in novel ways.

Contributions and Impact

- As a collaborating partner, my lab and I worked with scientists at KAUST and EPFL to provide visualization expertise and workflows, and reference configurations for both hardware and software for EPFL's own visualization facilities dedicated for the project. A letter of invite, as part of the collaboration with EPFL is provided here

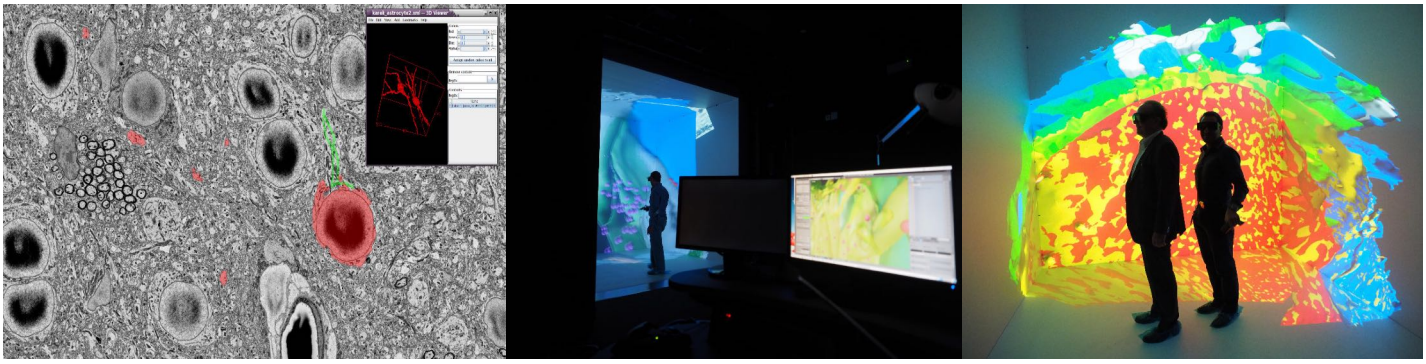


Figure 5: Reconstructing brain neurons (left) and visualizing them in my lab's CORNEA Virtual Reality Environment (center, and right)

- I hired the founding team that would provide the Visualization and 3D reconstruction expertise for the project at KAUST. The team was trained and incubated in my lab, before branching out and working full-time with Prof. Pierre Magistretti.
- Providing innovations in high-resolution 3D Reconstruction: For reconstructing 3D structures of neurons (see Figure 5), cross-sections of mouse brain tissue scanned under an electron microscope were reconstructed using special software, allowing a multidisciplinary team of researchers to examine the tissue in a 3D virtual environment in the lab. We created innovative improvements in 3D Reconstruction techniques to handle large imaging stacks (26GB). Such a stack would have required 1TB RAM and several weeks to months, to accurately reconstruct all features. My lab created workflows to perform 3D reconstruction on large image stacks in just a few hours, on a single workstation [Holst et al., 2016].
- Using the lab's high-end virtual-reality facilities: Due to the complexity of the reconstructed 3D models of brain tissues, qualitative observations are difficult even for an experienced neuroscientist. With the help of visualization and virtual reality facilities and software we developed in my lab, researchers could "walk" into a 3D model containing neuronal and astrocytic processes. Many qualitative and quantitative discoveries were enabled through this workflow [Calì et al., 2016] [Calì et al., 2017] [Agus et al., 2018]. For instance, the use of CORNEA Virtual Reality in the lab was key to the observation of a non-random distribution of glycogen, and led the team to develop tools to quantitatively analyze glycogen clustering and proximity to other sub-cellular features [Calì et al., 2016].

Publications

I am a co-author in the following publication:

- Three-dimensional immersive virtual reality for studying cellular compartments in 3D models from EM preparations of neural tissues [Calì et al., 2016]

My lab's resources have been used and acknowledged in the following publications:

- Adding large EM stack support [Holst et al., 2016]
- Visual Analysis of Glycogen Derived Lactate Absorption in Dense and Sparse Surface Reconstructions of Rodent Brain Structures [Calì et al., 2017]
- GLAM: Glycogen-derived Lactate Absorption Map for visual analysis of dense and sparse surface reconstructions of rodent brain structures on desktop systems and virtual environments [Agus et al., 2018]

Articles

- The Brain Gamers
- Exploring the Brain by Mimicking the Brain

Videos



Figure 6: Identifying and Visualizing new turbulence patterns in particle-driven gravity currents, on the lab's 40-megapixel tiled-display wall

- 3D Reconstruction of Hippocampus for visual analysis of brain structures
- Use of CORNEA Virtual Reality Environment for 3D Reconstructed Glia and Neuron Cells
- A scientific narrative produced by my lab, in collaboration with Dr. Magistretti's group

2.5 Visualizing Turbulence Patterns in Particle Driven Gravity Currents

Research Context

Particle driven currents simulate and study the flow of silt-filled water (i.e. rivers carrying sand, clay) into highly saline water bodies (oceans, seas). Such flows deposit large amounts of silt, clay and other sediments on the ocean floor over several years, and determine the topography and geology of the floor.

Contributions and Impact

I worked with the domain scientists to visualize new turbulence patterns that were discovered for the first time. Specifically, two interacting "instabilities" were discovered - Kelvin-Helmholtz instability (due to shear) and "fingering" instability (due to gravity). These interactions are quite complex and hard to observe experimentally. The instabilities were visualized and observed for the first-time ever through this work (see Figure 6).

These patterns determine how sediments are deposited on the sea-floor. Therefore, modeling, visualizing and analyzing this phenomena is critical for many reasons - for determining the laying of sub-marine communication cable pathways, for evaluating viable drilling regions of off-shore hydrocarbon deposits etc. For instance, an off-shore drilling operation can cost up to $200 million versus $10 million for on-shore drilling.

Publications

My lab's resources have been used and acknowledged in the following publications:

- An Analysis of Mesopicnal Flows: Intrusive Density- and Particle-Driven Gravity Currents via Direct Numerical Simulations [Espath et al., 2019]

Videos

- Dynamics of turbulent structures in particle-driven gravity currents

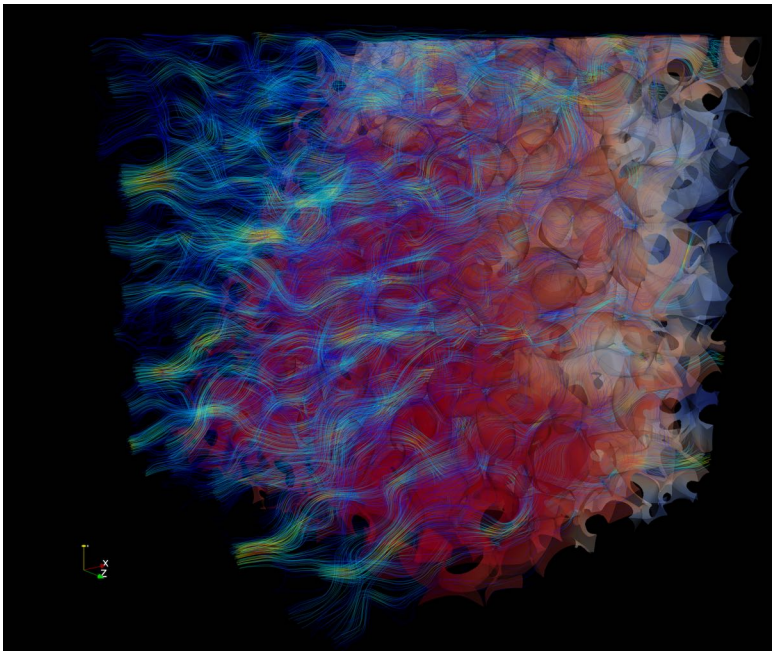


Figure 7: Observing flow-characteristics through a porous media.

2.6 Characterizing Fluid-Flow Through Porous Media

Research Context

Fluid flow through porous media is the manner in which fluids behave when flowing through a porous medium, for example sponge or wood, or when filtering water using sand or other porous material. The phenomena is also observed in geophysics, when extracting hydrocarbons from reservoirs, where oil and gas flows pores inside rocks.

Contributions and Impact

A detailed visualization that shows the phenomena of "flow" through a porous rock sample was implemented (see Figure 7). The geometry for the rock sample was obtained through Micro-CT - a technique that allows a non-destructive 3D imaging and reconstruction of objects. The streamlines show the flow of a fluid through the void-spaces in the sample, with colors depicting the speed of the flow (red is high velocity, blue is low velocity). Such visualization and analysis of flow models can be used to understand multi-phase flow (eg. oil, water) through rocks underground, for instance.

Publications

My lab's resources have been used and acknowledged in the following publications:

- Pore-scale simulation of fluid flow and solute dispersion in three-dimensional porous media [Icardi et al., 2014]

2.7 Other Contributions

The following list summarizes my contribution in other specific areas, as indicated:

- Implementation of hardware and software workflows for Virtual Reality facilities, and using them for training non-experts:
  - The future of the CAVE [DeFanti et al., 2011].
  - Acquisition of stereo panoramas for display in VR environments [Ainsworth et al., 2011].
  - Towards a digital makkah – Using immersive 3d environments to train and prepare pilgrims [Schneider et al., 2011].
- 3D Animation and Behavior modeling of 3D virtual characters, using Computer Graphics and Machine Learning:

- Semi-automatic End-user Tools for Construction of Virtual Avatar Behaviors [**Srinivasan, M.** and Metoyer, 2011]
- A Data-Driven Model of Pedestrian Movement [Casburn et al., 2007]
- Controllable Real-time Locomotion Using Mobility Maps [**Srinivasan, M.** et al., 2005]
- A responsiveness metric for controllable characters [**Srinivasan, M.** and Metoyer, 2005]
- A Tangible Interface for High-Level Direction of Multiple Animated Characters [Metoyer et al., 2003]

2.8 Indirect Contributions

The following contributions were enabled by staff and facilities from my lab. I have supervised, consulted with scientists, and provided inputs to each of these activities.

- Improving Pore Network Imaging & Characterization of Microporous Carbonate Rocks Using Multi-Scale Imaging Techniques: We supported 3D reconstruction of rock-core samples that were CT-Scanned. The reconstruction helps quantify pore characteristics in rock samples excavated from potential oil-drilling sites, which in turn allows the estimation of oil-reserves in the area [Hassan et al., 2017].
- Experimental and numerical characterization of the water flow in spacer-filled channels of spiral-wound membranes: We helped the researchers at KAUST's Water Desalination Research Center to visualize the flow characteristics of water over de-salination membranes. Our contributions help them understand how de-salination membranes degrade and fail, and to design efficient membranes that are cost-effective [Bucs et al., 2015].
- Visualization for Spatio-temporal Statistics: The latest advances in computing and virtual reality (VR) in my lab have enabled statistics researchers at KAUST to develop a suite of apps that allow users to visualize and interpret large and complex datasets in three dimensions.
  - Paper: Visuanimation in Statistics [Genton et al., 2015]
  - Paper: Visualizing spatiotemporal models with virtual reality: from fully immersive environments to applications in stereoscopic view [Castruccio et al., 2019]
  - Article: New stats apps show a virtual reality
- Data Science training program: With an increase in demand for the use of data science tools and workflows in scientific areas at KAUST, I provided a strategic direction for my lab to hire a Data Scientist (2018) and institute a strong data-science program for training and research.
  - Article: Introduction to Data Science Workshops - Learning the tools of the trade.

3 Awards

Leading@KAUST Program by Harvard Business Publishing — KAUST, Saudi Arabia
Nominated Participant — March 2015

The Leading@KAUST Management program administered by Harvard Business Publishing is an exclusive program for nominated high-performing staff that follows a blended learning approach for those transitioning into a leadership role. A group of 25 distinguished staff were selected for a 10-week program. The program elements are designed to

- Ensure a smooth transitional experience to leadership roles
- Enforce the leadership and managerial skills of our managers to lead and coach their teams effectively
- Enhance existing knowledge and acquire new, in relation to contemporary management best practices

Recognition for Distinguished Service — KAUST, Saudi Arabia
Five-year Service Award — May 2014

Presented by the President of the University, in recognition and appreciation of five years of loyal and distinguished service with King Abdullah University of Science and Technology. About 150 staff from about 3000 staff were selected for this award.

4 Student Supervision and Advising

Ph.D. Thesis: Mr. T. Muralidharan — MGR Research Institute, India
Hybrid Extensive Machine Learning Algorithm For Domain Specific Features To Detect And Classify Web Spam — March 2019

Abstract: Machine learning deals with making machines learn by experience. It involves data science, artificial intelligence and statistics. It is one of today's most rapidly growing technical fields. This thesis presents a Hybrid Extensive Machine Learning Algorithm (HEMLA) for classification and clustering techniques in data mining. Recent progress in machine learning has been driven both by the development of new learning algorithms and theory and by the ongoing explosion in the availability of online data and low-cost computation. Through experimental study it is found that the proposed Hybrid Extensive Machine Learning Algorithm outperforms the existing supervised and unsupervised machine learning algorithms for detecting and classifying Web Spam.

Master's Thesis: Ms. Omniah Nagoor — KAUST, Saudi Arabia
Image-based Exploration of Large-Scale Pathline Fields — May 2014

Abstract: *PathlinesExplorer* is a novel image-based tool, which has been designed to visualize large-scale pathline fields on a single computer. PathlinesExplorer integrates explorable images (EI) technique with order-independent transparency (OIT) method. The method is unique in that it allows users to handle large data on a single workstation. Although it is a view-dependent method, *PathlinesExplorer* combines both exploration and modification of visual aspects without re-accessing the original huge data. Our approach is based on constructing a per-pixel linked list data structure in which each pixel contains a list of pathline segments. With this view-dependent method, it is possible to filter, color-code, and explore large-scale flow data in real-time. In addition, optimization techniques such as early-ray termination and deferred shading are applied, which further improves the performance and scalability of this approach.

Master's Thesis: Ms. Roba Binyahib — KAUST, Saudi Arabia
Image-based Exploration of Iso-surfaces for Large Multi-Variable Datasets using Parameter Space — May 2013

Abstract: Explorable Images (EI) are a novel approach that provides limited interactive visualization without the need to re-render from the original data. In this work, the concept of EI has been used to create a workflow that deals with explorable iso-surfaces for scalar fields in a multivariate, time-varying dataset. This work presents a tool that allows a user, at run-time, to interactively browse and calculate a combination of iso-surfaces superimposed on each other. The result is the same as calculating multiple iso-surfaces from the original data but without the memory and processing overhead. The tool also allows the user to change the (scalar) values superimposed on each of the surfaces, modify their color map, and interactively re-light the surfaces. The work demonstrates the effectiveness of this approach over a multi-terabyte combustion dataset. It also illustrates the efficiency and accuracy of the technique by comparing results with those from a more traditional visualization pipeline.

5 Program Committee Service

I have served in the following conferences, workshops and symposia as a program committee member, co-organizer or as a planning committee member. A brief outline about each of the event is provided below.

2019	Program Committee, EuroGraphics Symposium on Parallel Graphics and Visualization	Porto, Portugal
2018	Program Committee, EuroGraphics Symposium on Parallel Graphics and Visualization	Brno, Czech
2017	Program Committee, EuroGraphics Symposium on Parallel Graphics and Visualization	Barcelona, Spain
2016	Program Committee, IEEE Symposium on Large Data Analysis and Visualization	Baltimore, MD
2014	Program Committee, IEEE Symposium on Large Data Analysis and Visualization	Paris, France
2013	Program Committee, IEEE Symposium on Large Data Analysis and Visualization	Atlanta, GA
2017	Organizing Committee, High Performance Computing - Saudi Arabia	Saudi Arabia
2015	Planning Committee, Supercomputing Conference	Austin, TX
2014	Planning Committee, Supercomputing Conference	New Orleans, LA
2015	Co-Organizer, Workshop on Visualization Infrastructure and Systems Technology	Austin, TX

EuroGraphics Symposium on Parallel Graphics and Visualization: The aim of this symposium is to foster the exchange of experiences and knowledge exploiting and defining new trends in parallel graphics and visualization. The importance of parallel computing is increasing rapidly with the ubiquitous availability of multi-core CPUs, GPUs, and cluster systems. Computationally demanding and data-intensive applications in graphics and visualization are strongly affected by this trend and require novel, efficient parallel solutions.

IEEE Symposium on Large Data Analysis and Visualization: Modern large-scale scientific simulations, sensor networks, and experiments are generating enormous datasets, with some projects approaching the multiple exabyte range in the near term. Managing and analyzing large datasets in order to transform it into insight is critical for a variety of disciplines including climate science, nuclear physics, security, materials design, transportation, and urban planning. The tools and approaches needed to mine, analyze, and visualize data at extreme scales can be fully realized only with end-to-end solutions, which demands collective, interdisciplinary efforts. The LDAV symposium brings together domain scientists, data analysts, visualization researchers, users, designers and artists, to foster common ground for solving both near and long-term problems.

High Performance Computing Saudi Arabia: High Performance Computing Saudi Arabia (HPC Saudi) is the premier regional event in the field, where participants can meet each other, share ideas and experiences, and discuss cooperation and collaboration. The event focuses on coordinated efforts for the advancement of the HPC ecosystem in the Kingdom.

Supercomputing Conference Series: The Supercomputing Conference prides itself as one of the most competitive and valued technical conferences in the field of High Performance Computing. It is often listed at the very top of various national conference ranking systems, showcasing the impact that publishing at SC has on the visibility and careers of researchers around the world. Research published at SC often has amongst the highest citation counts and is regularly considered to be the most impactful research in the community.

Visualization Infrastructure & Systems Technology: The workshop is organized as part of the Supercomputing Conference, and explores the intersection between human perception and large-scale visual analysis through the study of visualization interfaces and interactive displays. This rich intersection includes virtual reality systems, visualization through augmented reality, large-scale visualization systems, novel visualization interfaces, high-resolution interfaces, mobile displays, and visualization display middleware. The workshop provides a forum in which experts in the large-scale visualization technology field and users can come together to discuss state-of-the art technologies for visualization infrastructures and systems.

6 Other Recognitions

Dagstuhl Seminar: *In Situ* Visualization for Computational Science — Dagstuhl, Germany
Invited Contributor & Participant — July 2018

- Group Co-Leader: *Workflow Specification Group* [Parashar et al., 2019]
- Technical Talk: *Supporting the Virtual Red Sea Project at KAUST*
- Group Contributor:

- *Exascale Requires Different Data* [Bauer et al., 2019]
- *Software Complexity, Heterogeneity, and User-facing Issues* [Bethel et al., 2019]

Schloss Dagstuhl, a nonprofit institution with associates from eleven universities and research organizations, is one of the world's leading research centers in informatics. It has been hosting invitation-only Dagstuhl Seminars and Dagstuhl Perspectives Workshops since the center's founding in 1990. The seminar workshop on *In Situ* Visualization for Computational Science **hosted 35 world-leading experts to identify ten challenges for in-situ visualization and processing that require significant research.**

In Situ Visualization, i.e., visualizing simulation data as it is generated, is an emerging processing paradigm in response to trends in the area of high-performance computing. This paradigm holds great promise in its ability to access increased spatio-temporal resolution and leverage extensive computational power. However, the paradigm is also widely viewed as limiting when it comes to exploration-oriented use cases and further will require visualization systems to become more and more complicated and constrained. Additionally, there are many open research topics with *in situ* visualization. The seminar brought together researchers and practitioners from three communities (computational science, high-performance computing, and scientific visualization) to share interesting findings, to identify lines of open research, and to determine a medium-term research agenda that addresses the most pressing problems. These challenges were identified by spending the first day of the workshop with participants giving short presentations on their experiences with in-situ processing, with a special focus on unsolved problems. The participant perspectives were then organized into the ten research challenges. Over the following days, sub-groups discussed each of the ten challenges and then presented the key points of their discussions to the group and received feedback. Shortly after the workshop, the leaders of each sub-group wrote summaries for its associated research challenge; these summaries are the basis of the report In Situ Visualization for Computational Science (Dagstuhl Seminar 18271), Volume 8, Issue 7, July 2018.

Workshop on Visualization Infrastructure & Systems Technology — New Orleans, U.S.A
Invited Speaker: KAUST Visualization Laboratory - Sustaining and Operating at Scale — November 2014

The VISTech (Visualization Infrastructure & Systems Technology) workshop is organized as part of the Supercomputing Conference, and explores the intersection between human perception and large-scale visual analysis through the study of visualization interfaces and interactive displays. This rich intersection includes virtual reality systems, visualization through augmented reality, large-scale visualization systems, novel visualization interfaces, high-resolution interfaces, mobile displays, and visualization display middleware. The workshop provides a forum in which experts in the large-scale visualization technology field and users can come together to discuss state-of-the art technologies for visualization infrastructures and systems.

Microsoft Research eScience Workshop — Stockholm, Sweden
Invited Panel Speaker: *Is NUI "Natural" for Scientists?* — December 2011

Each year, 150 top researchers join with Microsoft Research to examine the successes and opportunities for research in the field of data-intensive science. The goal of this eighth annual cross-disciplinary workshop is to bring together scientists from diverse disciplines to share their research and discuss how computing is transforming their work.

Natural user interface (NUI) technology has captured imaginations across industry, education, and entertainment with the release of groundbreaking new technologies for interacting with computer systems, such as Microsoft Surface and Kinect for Xbox 360. Technology is becoming more natural and intuitive - people use touch and speech to interact with their phones, at the ATM, at the grocery store, and in their cars. The learning curve for working with computers is becoming less and less of a barrier, thanks to more natural ways to interact. How is this technology being used by scientists today, and what is its potential? This panel explores best practices and new directions, ultimately seeking to present a slice through the current research agenda for NUI research in scientific applications.

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